Write Once, Accelerate Everywhere: GPU-Ready Java with TornadoVM

Thanos Stratikopoulos
Research Fellow at the University of Manchester

February 6th, 2026
Voxxed Days Ticino 2026

About me

Dr Thanos Stratikopoulos
Research Fellow and Impact Champion
The University of Manchester

athanasios.stratikopoulos@manchester.ac.uk
@thanos_str
https://www.linkedin.com/in/stratika/
@stratika.bsky.social

Agenda

MANCHESTER 1824
The University of Manchester

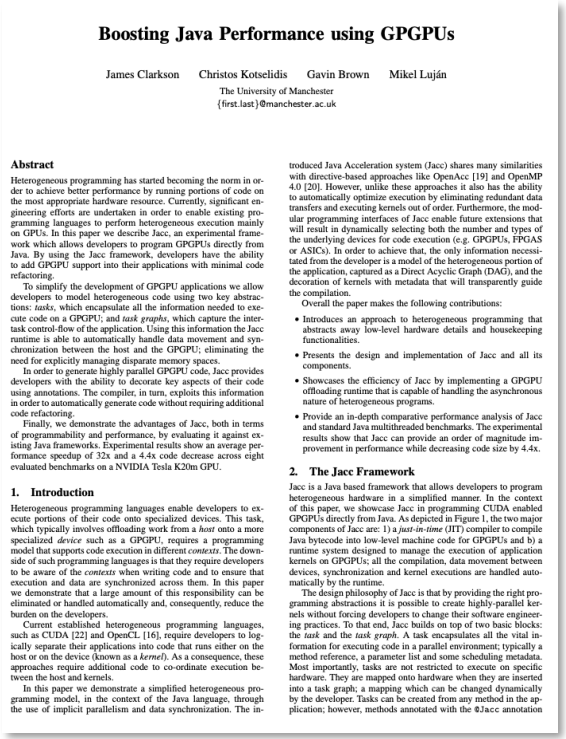
- Brief History of TornadoVM
- TornadoVM Overview
- TornadoVM Types with Foreign Function & Memory API (Project Panama)
- Setup TornadoVM
- Integration in Java Projects
- More Complex Cases
- Summary

TornadoVM History

How it started…

- PhD research project 2013, Uni. Manchester
- Initially named Jacc it later evolved to TornadoVM
- Originally based on SOOT framework
- Designed to understand how the JVM could tackle GPUs

MANCHESTER 1824
The University of Manchester



Boosting Java Performance using GPGPUs

James Clarkson Christos Kotselidis Gavin Brown Mikel Luján
The University of Manchester
{first.last}@manchester.ac.uk

Abstract

Heterogeneous programming has started becoming the norm in order to achieve better performance by running portions of code on the most appropriate hardware resource. Currently, significant engineering efforts are undertaken in order to enable existing programming languages to perform heterogeneous execution mainly on GPUs. In this paper we describe Jacc, an experimental framework which allows developers to program GPGPUs directly from Java. By using the Jacc framework, developers have the ability to add GPGPU support into their applications with minimal code refactoring.

To simplify the development of GPGPU applications we allow developers to model heterogeneous code using two key abstractions: *tasks*, which encapsulate all the information needed to execute code on a GPGPU; and *task graphs*, which capture the inter-task control-flow of the application. Using this information the Jacc runtime is able to automatically handle data movement and synchronization between the host and the GPGPU; eliminating the need for explicitly managing disparate memory spaces.

In order to generate highly parallel GPGPU code, Jacc provides developers with the ability to decorate key aspects of their code using annotations. The compiler, in turn, exploits this information in order to automatically generate code without requiring additional code refactoring.

Finally, we demonstrate the advantages of Jacc, both in terms of programmability and performance, by evaluating it against existing Java frameworks. Experimental results show an average performance speedup of 32x and a 4.4x code decrease across eight evaluated benchmarks on a NVIDIA Tesla K20m GPU.

1. Introduction

Heterogeneous programming languages enable developers to execute portions of their code onto specialized devices. This task, which typically involves offloading work from a *host* onto a more specialized *device* such as a GPGPU, requires a programming model that supports code execution in different *contexts*. The downside of such programming languages is that they require developers to be aware of the *contexts* when writing code and to ensure that execution and data are synchronized across them. In this paper we demonstrate that a large amount of this responsibility can be eliminated or handled automatically and, consequently, reduce the burden on the developers.

Current established heterogeneous programming languages, such as CUDA [22] and OpenCL [16], require developers to logically separate their applications into code that runs either on the host or on the device (known as a *kernel*). As a consequence, these approaches require additional code to co-ordinate execution between the host and kernels.

In this paper we demonstrate a simplified heterogeneous programming model, in the context of the Java language, through the use of implicit parallelism and data synchronization. The introduced Java Acceleration system (Jacc) shares many similarities with directive-based approaches like OpenAcc [19] and OpenMP 4.0 [20]. However, unlike these approaches it also has the ability to automatically optimize execution by eliminating redundant data transfers and executing kernels out of order. Furthermore, the modular programming interfaces of Jacc enable future extensions that will result in dynamically selecting both the number and types of the underlying devices for code execution (e.g. GPGPUs, FPGAS or ASICs). In order to achieve that, the only information necessitated from the developer is a model of the heterogeneous portion of the application, captured as a Direct Acyclic Graph (DAG), and the decoration of kernels with metadata that will transparently guide the compilation.

Overall the paper makes the following contributions:

- Introduces an approach to heterogeneous programming that abstracts away low-level hardware details and housekeeping functionalities.
- Presents the design and implementation of Jacc and all its components.
- Showcases the efficiency of Jacc by implementing a GPGPU offloading runtime that is capable of handling the asynchronous nature of heterogeneous programs.
- Provide an in-depth comparative performance analysis of Jacc and standard Java multithreaded benchmarks. The experimental results show that Jacc can provide an order of magnitude improvement in performance while decreasing code size by 4.4x.

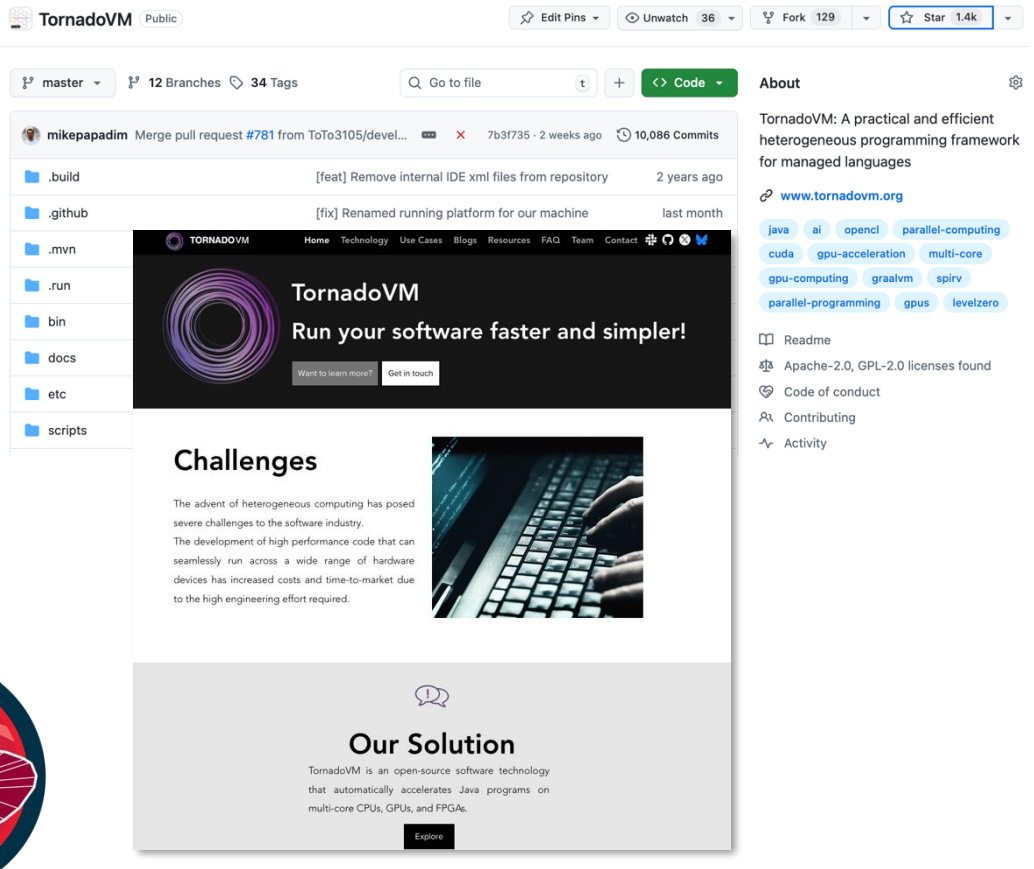
2. The Jacc Framework

Jacc is a Java based framework that allows developers to program heterogeneous hardware in a simplified manner. In the context of this paper, we showcase Jacc in programming CUDA enabled GPGPUs directly from Java. As depicted in Figure 1, the two major components of Jacc are: 1) a *just-in-time* (JIT) compiler to compile Java bytecode into low-level machine code for GPGPUs and b) a runtime system designed to manage the execution of application kernels on GPGPUs; all the compilation, data movement between devices, synchronization and kernel executions are handled automatically by the runtime.

The design philosophy of Jacc is that by providing the right programming abstractions it is possible to create highly-parallel kernels without forcing developers to change their software engineering practices. To that end, Jacc builds on top of two basic blocks: the *task* and the *task graph*. A task encapsulates all the vital information for executing code in a parallel environment; typically a method reference, a parameter list and some scheduling metadata. Most importantly, tasks are not restricted to execute on specific hardware. They are mapped onto hardware when they are inserted into a task graph; a mapping which can be changed dynamically by the developer. Tasks can be created from any method in the application; however, methods annotated with the `@Jacc` annotation

How it is going…

- Open-sourced in 2018
- Transitioned to Graal
- 23 releases; currently at v2.2.0
- 30 contributors
- Production deployment at ESA's GAIA mission



Funded by the European Union (AERO: 101092850) & UKRI under the Horizon Europe funding guarantee (AERO: 10048318).

https://aero-project.eu/

TornadoVM Mature Ecosystem

JDK 21 Distributions

- OpenJDK
- GraalVM
- Eclipse Temurin
- RedHat Mandrel
- Windows JDK
- Azul Zulu
- SapMachine

Truffle Framework

Java, JS, Python, R, Ruby, WebAssembly

GraalVM

Plugin to JDK

TORNADOVM

Hardware Backends

- OpenCL
- NVIDIA CUDA
- SPIR
- (soon)

Docker Images

beehivelab/tornadovm-nvidia-openjdk
By beehivelab · Updated 2 days ago

beehivelab/tornadovm-intel-openjdk
By beehivelab · Updated 2 days ago
TornadoVM Docker Image for Intel Platforms: It includes the OpenCL and the SPIR-V Backend.

NVIDIA, intel

Developer Tools

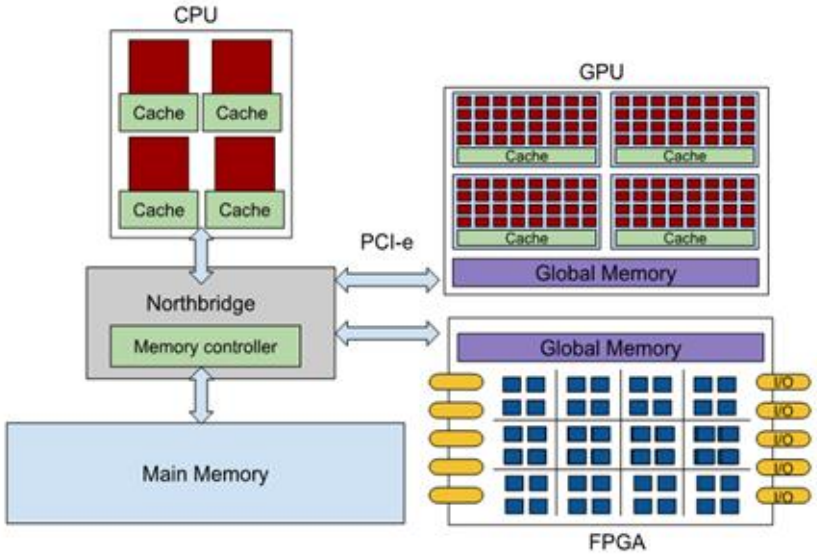
- TornadoVMPulse
- TornadoViz
- TornadoInsight
- NVIDIA Nsight Systems

OpenJDK25
GraalVM25
(*soon*)

Integrations

How much different is GPU programming?

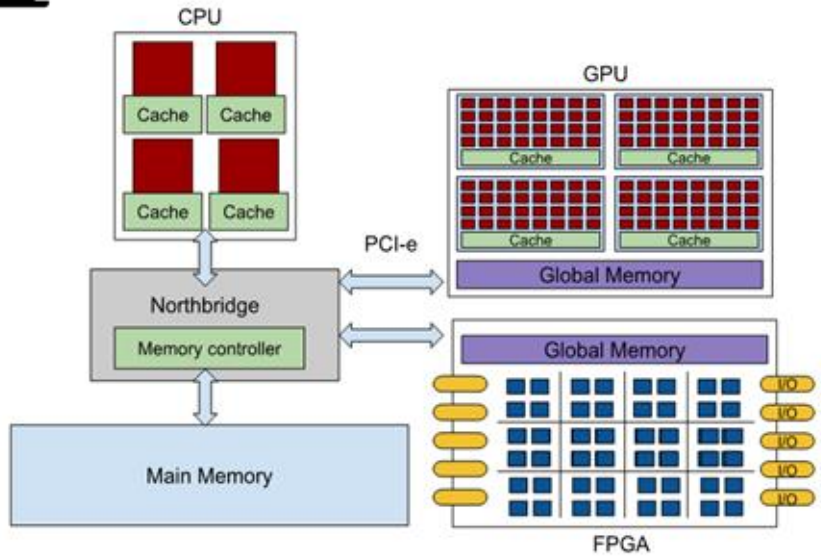
Very different!



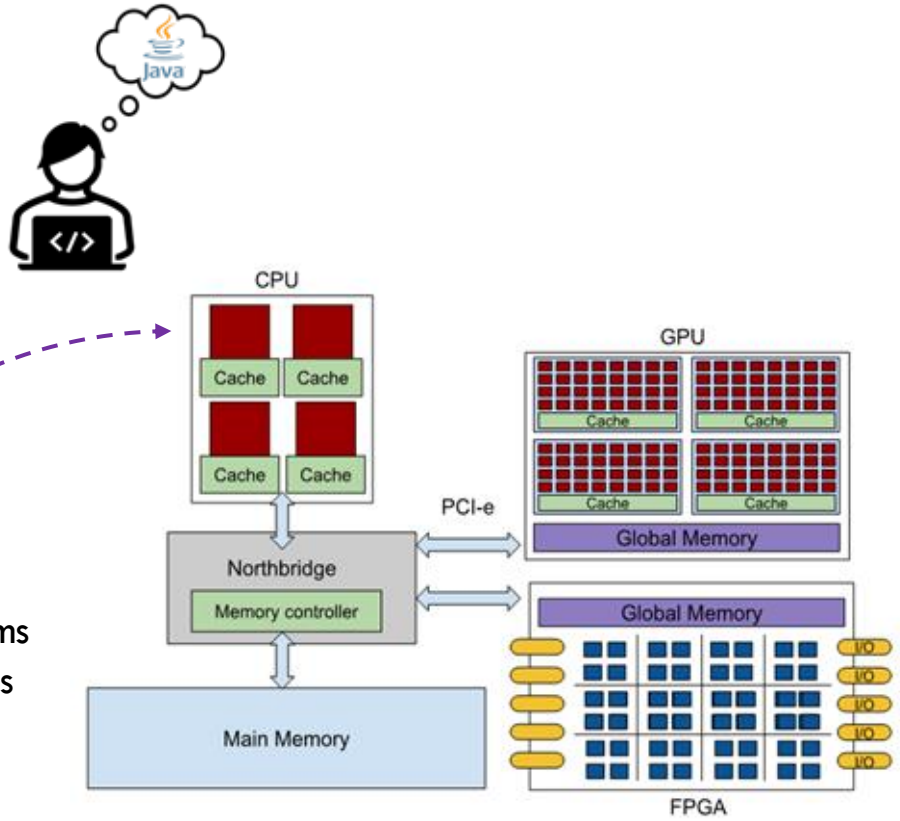
Very different!

Parallel APIs

- Vector API
- Parallel Streams
- Virtual Threads



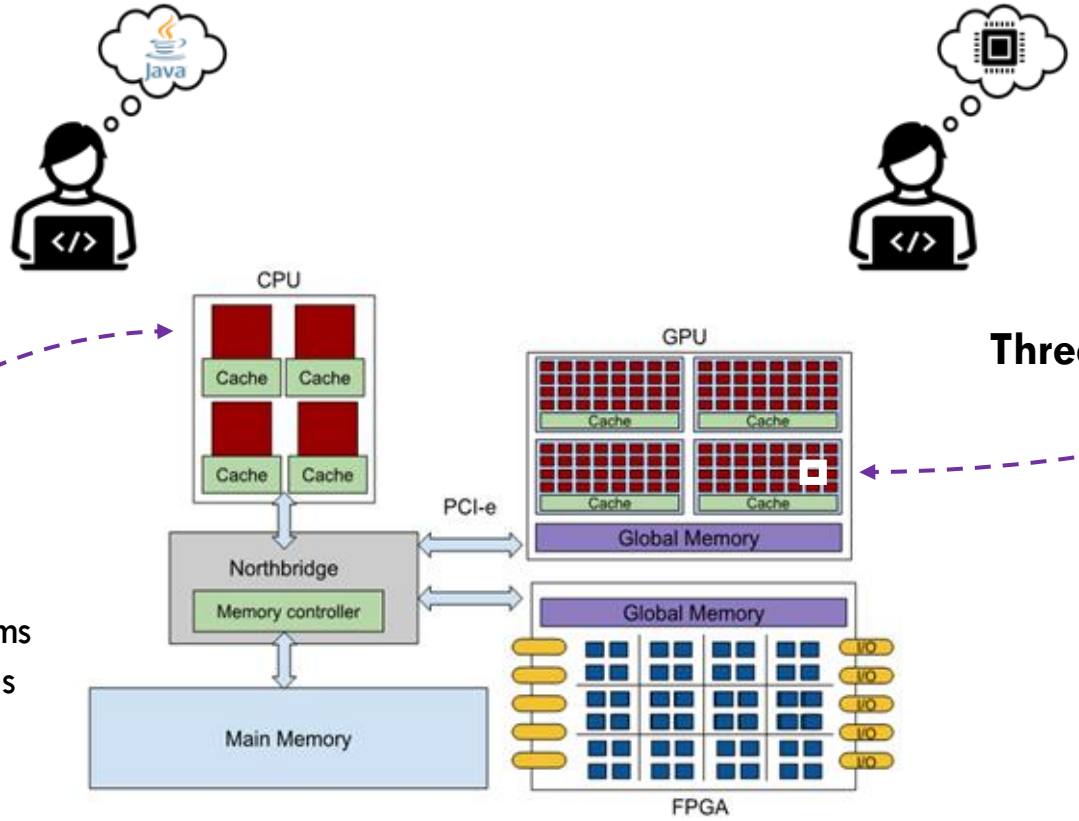
Very different!



Parallel APIs

- Vector API
- Parallel Streams
- Virtual Threads

Very different!

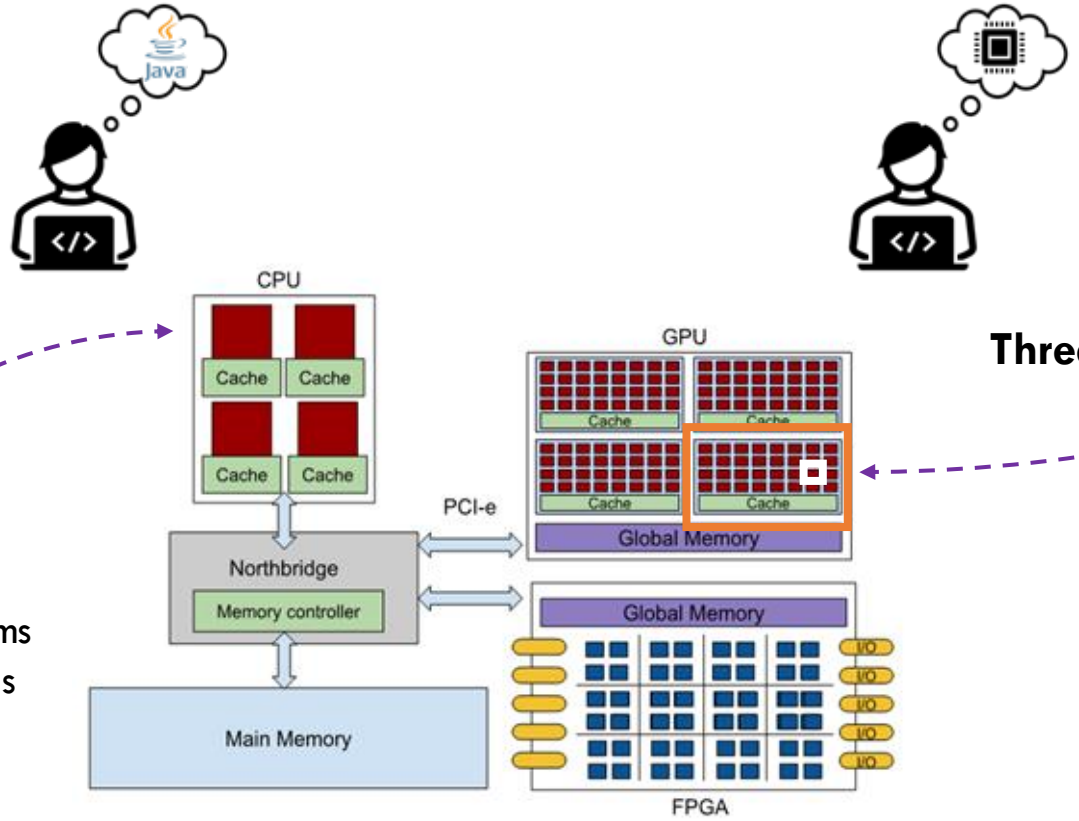


Parallel APIs

- Vector API
- Parallel Streams
- Virtual Threads

Thread-level computation

Very different!



Parallel APIs

- Vector API
- Parallel Streams
- Virtual Threads

Thread-level computation

Key remarks regarding GPU programming

- GPU programming requires a *different mindset* than programming in Java
- Programmers must be familiar with thread-level programming and grids
- Lots of verbosity, since developers **must** create explicitly all data structures

It can break the abstraction! Remember Java's promise:
Write Once, Run Anywhere!

TornadoVM Overview

@tornadovm

@tornadovm.org

info@tornadovm.org

https://tornadovm.org

https://github.com/beehive-lab/TornadoVM

SCAN ME

A JVM plugin that accelerates Java methods on heterogeneous hardware!

Features:

- Open source
- Platform agnostic
- Automatic code optimization
- Off-heap data types (Foreign Function & Memory API)

TornadoVM Types with Foreign Function & Memory API (Project Panama)

Objects Allocated on Heap

MANCHESTER 1824
The University of Manchester

```
public static void main(String[] args) {
    int size = 512;

    float[] matrixA = new float[size * size];
    float[] matrixB = new float[size * size];
    float[] matrixC = new float[size * size];
    float[] resultSeq = new float[size * size];

    Random r = new Random();
    IntStream.range(0, size * size).parallel().forEach(idx -> {
        matrixA[idx] = r.nextFloat();
        matrixB[idx] = r.nextFloat();
    });

    matrixMultiplication(matrixA, matrixB, resultSeq, size);
```

Allocated on-heap.
So pruned to GC!

Using Off-heap Data

The University of Manchester

```java
public static void main(String[] args) {
    int size = 512;

    float[] matrixA = new float[size * size];
    float[] matrixB = new float[size * size];
    float[] matrixC = new float[size * size];
    float[] resultSeq = new float[size * size];

    Random r = new Random();
    IntStream.range(0, size * size).parallel().forEach(idx -> {
        matrixA[idx] = r.nextFloat();
        matrixB[idx] = r.nextFloat();
    });

    matrixMultiplication(matrixA, matrixB, resultSeq, size);
```

```java
public static void main(String[] args) throws Exception {
    int size = 512;

    FloatArray matrixA = new FloatArray( numberOfElements: size * size);
    FloatArray matrixB = new FloatArray( numberOfElements: size * size);
    FloatArray matrixC = new FloatArray( numberOfElements: size * size);
    FloatArray resultSeq = new FloatArray( numberOfElements: size * size);

    Random r = new Random();
    IntStream.range(0, size * size).parallel().forEach(idx -> {
        matrixA.set(idx, r.nextFloat());
        matrixB.set(idx, r.nextFloat());
    });

    matrixMultiplication(matrixA, matrixB, resultSeq, size);
```

https://tornadovm.readthedocs.io/en/latest/offheap-types.html#migrating-from-v0-15-2-to-v1-0

Utilizing TornadoVM Off-heap DataTypes

```
// allocate an off-heap memory segment that will contain 16 int values
IntArray intArray = new IntArray(16);
```

Additionally, developers can create an instance of a TornadoVM native array by invoking factory methods for different data representations. In the following examples we will demonstrate the API functions for the `FloatArray` type, but the same methods apply for all support native array types.

```
// from on-heap array to TornadoVM native array
public static FloatArray fromArray(float[] values);
// from individual items to TornadoVM native array
public static FloatArray fromElements(float... values);
// from Memory Segment to TornadoVM native array
public static FloatArray fromSegment(MemorySegment segment);
```

The main methods that the off-heap types expose to manage the Memory Segment of each type are presented in the list below.

```
public void set(int index, float value) // sets a value at a specific index
   E.g.:
       FloatArray floatArray = new FloatArray(16);
       floatArray.set(0, 10.0f); // at index 0 the value is 10.0f
public float get(int index) // returns the value of a specific index
   E.g.:
       FloatArray floatArray = FloatArray.fromArray(new float[] {2.0f, 1.0f, 2.0f, 5.0f});
       float floatValue = floatArray.get(3); // returns 5.0f
public void clear() // sets the values of the segment to 0
   E.g.:
       FloatArray floatArray = new FloatArray(1024);
       floatArray.clear(); // the floatArray contains 1024 zeros
public void init(float value) // initializes the segment with a specific value
   E.g.:
       FloatArray floatArray = new FloatArray(1024);
       floatArray.init(1.0f); // the floatArray contains 1024 ones
public int getSize() // returns the number of elements in the segment
   E.g.:
       FloatArray floatArray = new FloatArray(16);
       int size = floatArray.getSize(); // returns 16
public float[] toHeapArray(); // Converts the data from off-heap to on-heap
public long getNumBytesOfSegment(); // Returns the total number of bytes the underlying Memory
public long getNumBytesWithoutHeader(); // Returns the total number of bytes the underlying Me
```

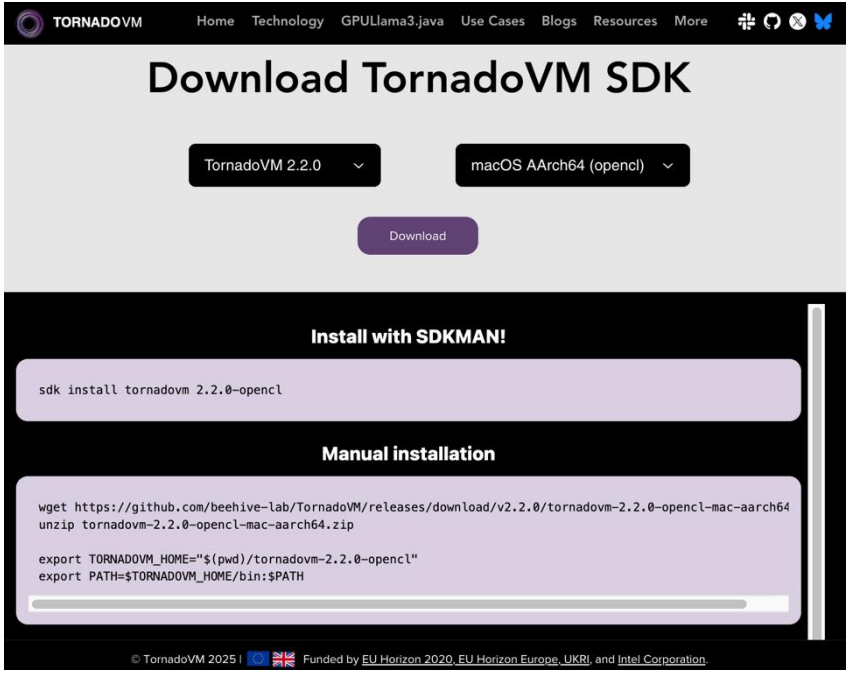
Factory methods to migrate your **data** to off-heap data types!

Factory method to migrate off-heap **data** to on-heap data types!

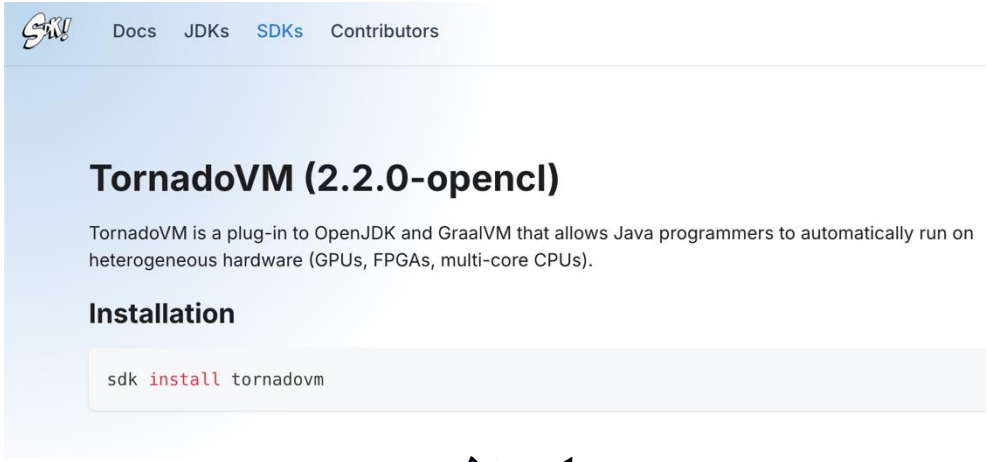
https://tornadovm.readthedocs.io/en/latest/offheap-types.html#migrating-from-v0-15-2-to-v1-0

Setup TornadoVM

SDK archives are shipping!



https://www.tornadovm.org/downloads



https://sdkman.io/sdks/tornadovm/

How to Install TornadoVM?

The University of Manchester

- Ensure you have a valid JDK distribution (e.g. Temurin):

```
sdk install java 21.0.9-tem
```

→ `$JAVA_HOME`

- Install TornadoVM SDK:

```
sdk install tornadovm 2.2.0-opencl
```

→ `$TORNADOVM_HOME`

```
tornado --devices

tornado --version
```

What is the tornado command?

```
$ tornado –cp MyApp.jar MyApp

$ java @$TORNADOVM_HOME/tornado-argfile –cp MyApp.jar MyApp
```

What is the tornado command?

```
$ tornado -cp MyApp.jar MyApp

$ java @$TORNADOVM_HOME/tornado-argfile -cp MyApp.jar MyApp
```

What is the tornado command?

```
$ tornado -cp MyApp.jar MyApp
```

```
$ java @$TORNADOVM_HOME/tornado-argfile -cp MyApp.jar MyApp
```

Contains the JVM Flags that enable TornadoVM:

```
tornado --printJavaFlags
```

```
-server
-XX:+UnlockExperimentalVMOptions
-XX:+EnableJVMCI
--enable-preview
-Djava.library.path=/Users/thanos/.sdkman/candidates/tornadovm/current/lib
--module-path .:/Users/thanos/.sdkman/candidates/tornadovm/current/share/java/tornado
-Dtornado.load.api.implementation=uk.ac.manchester.tornado.runtime.tasks.TornadoTaskGraph
-Dtornado.load.runtime.implementation=uk.ac.manchester.tornado.runtime.TornadoCoreRuntime
-Dtornado.load.tornado.implementation=uk.ac.manchester.tornado.runtime.common.Tornado
-Dtornado.load.annotation.implementation=uk.ac.manchester.tornado.annotation.ASMClassVisitor
-Dtornado.load.annotation.parallel=uk.ac.manchester.tornado.api.annotations.Parallel  --upgrade-module-path
/Users/thanos/.sdkman/candidates/tornadovm/current/share/java/graalJars
-XX:+UseParallelGC  @/Users/thanos/.sdkman/candidates/tornadovm/current/etc/exportLists/common-exports
@/Users/thanos/.sdkman/candidates/tornadovm/current/etc/exportLists/opencl-exports --add-modules ALL-
SYSTEM,tornado.runtime,tornado.annotation,tornado.drivers.common,tornado.drivers.opencl
```

What is the tornado command?

```
$ tornado -cp MyApp.jar MyApp
$ java @$TORNADOVM_HOME/tornado-argfile -cp MyApp.jar MyApp
```

Contains the JVM Flags that enable TornadoVM:

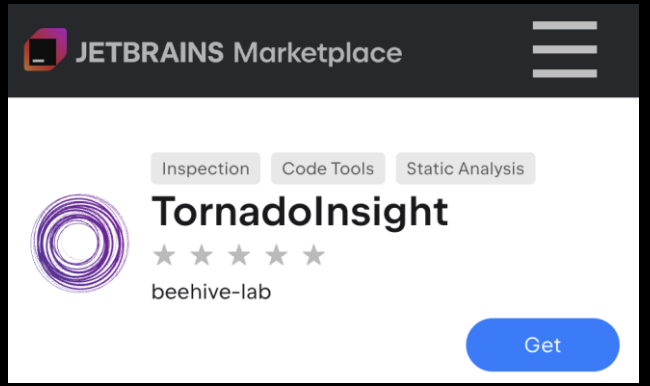
```
tornado --printJavaFlags
```

```
$ java @$TORNADOVM_HOME/tornado-argfile -cp $TORNADOVM_HOME/share/java/tornado/tornado-examples-2.2.0.jar uk.ac.manchester.tornado.examples.compute.MatrixVectorRowMajor
```

A TornadoVM Example of MatrixMultiplication

https://github.com/beehive-lab/TornadoVM

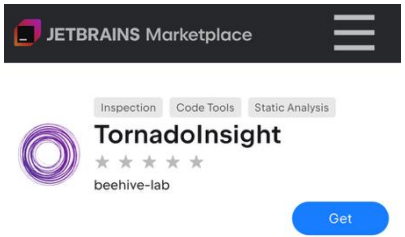
Branch: ***tornado-insight/demos***



IntelliJ & Plugin configuration

- Ensure you have installed JetBrains IntelliJ IDEA & TornadoInsight (1.5.0)

https://plugins.jetbrains.com/plugin/23309-tornadoinsight



https://p2code-project.eu/

Funded by the European Union (P2CODE: 101093069) & UKRI under the Horizon Europe funding guarantee (P2CODE: 10048316).

IntelliJ & Plugin configuration

- Ensure

https://pluc

TORNADOVM Home Technology GPULlama3.java Use Cases Blogs Resources More

Thanos Stratikopoulos · Dec 23, 2025 · 2 min read

TornadoInsight - Compatibility with TornadoVM SDK 2.0+ & Configuration Guide

This blog updates the previously published **TornadoInsight** configuration guidelines and explains how to configure the required environment variable to ensure that TornadoInsight correctly detects the TornadoVM SDK when IntelliJ IDEA is launched from a graphical environment.

Overview

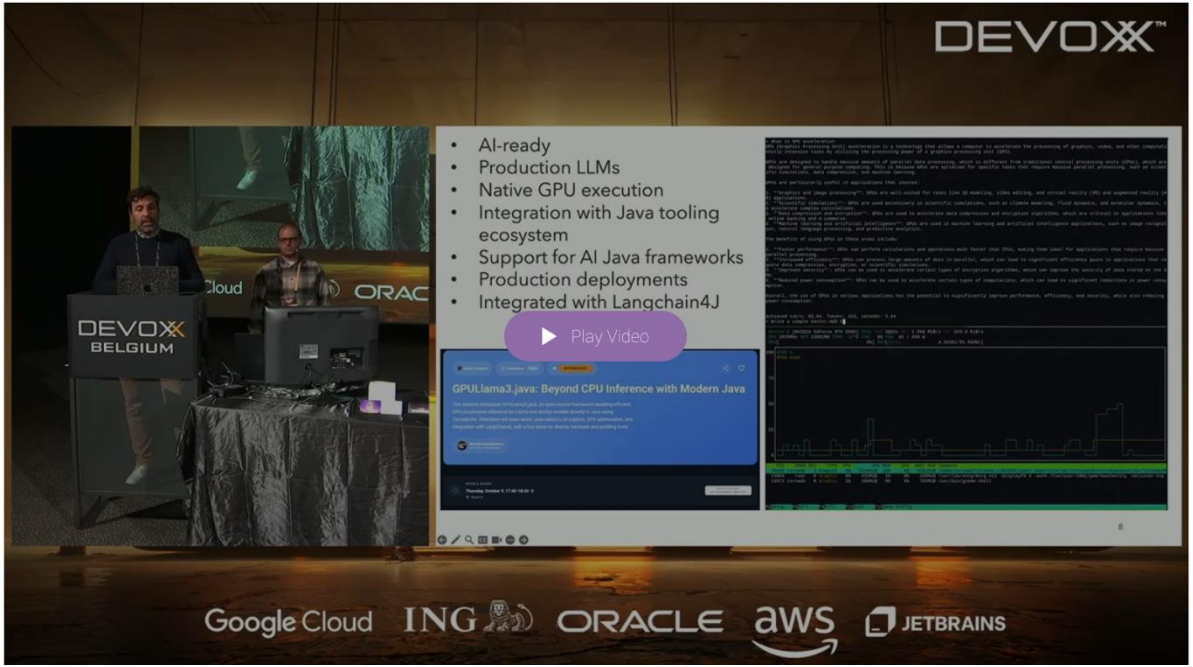
TornadoInsight requires access to a compatible Java Development Kit (JDK) and the TornadoVM SDK. We explained in previous blogs the guidelines for installing the TornadoVM SDKs (manually) or via SDKMAN!.

https://tornadovm.org/blog

TornadoVM Deep-dive

MANCHESTER 1824
The University of Manchester

DevoxxBE 2025



Integration in Java Projects

How to use TornadoVM in my project?

MANCHESTER 1824
The University of Manchester

- Jars pushed in Maven Central

Maven™

Integrating TornadoVM in your project

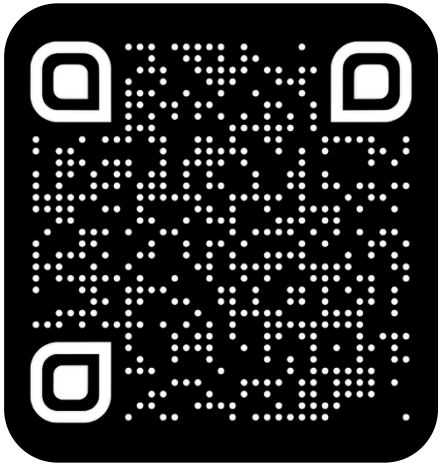
For Maven projects (pom.xml)

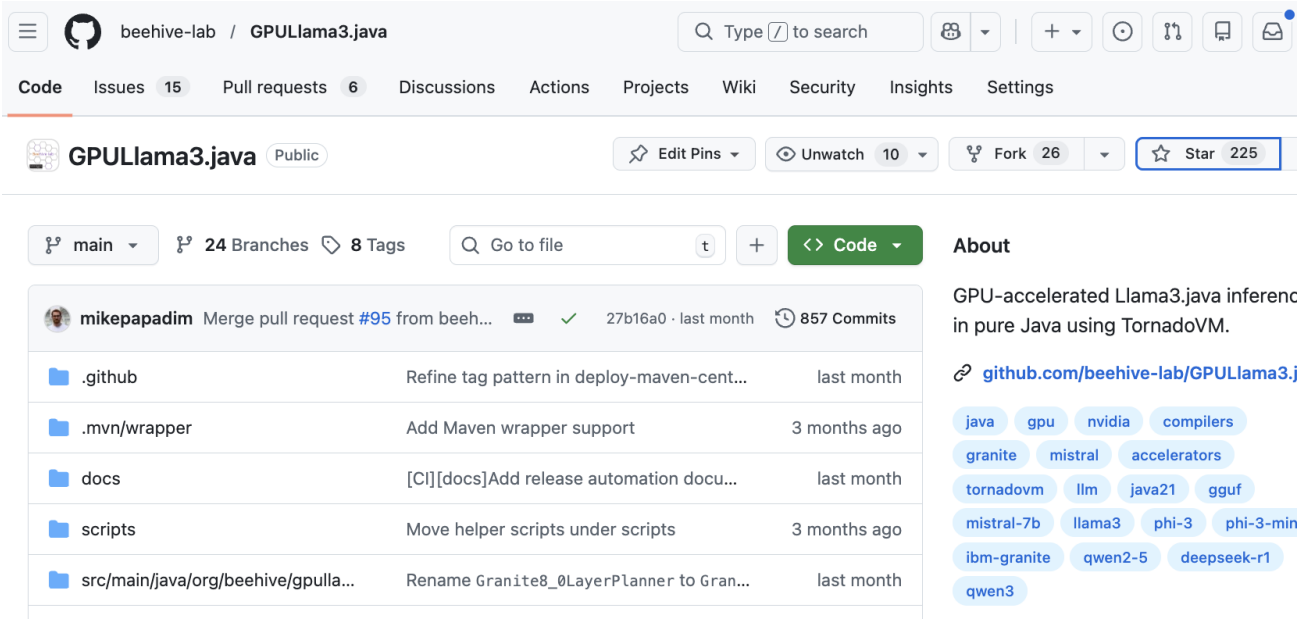
```
<dependencies>
  <dependency>
    <groupId>io.github.beehive-lab</groupId>
    <artifactId>tornado-api</artifactId>
    <version>2.2.0</version>
  </dependency>
  <dependency>
    <groupId>io.github.beehive-lab</groupId>
    <artifactId>tornado-runtime</artifactId>
    <version>2.2.0</version>
  </dependency>
</dependencies>
```

More Complex Cases (GPULlama3.java)

GPULlama3.java

The University of Manchester





https://github.com/beehive-lab/GPULlama3.java

About GPULlama3.java

MANCHESTER 1824
The University of Manchester

- Java-based GPU capable LLM Inference Engine
  - built on top of llama3.java for model ingestion
  - built with TornadoVM for GPU acceleration
- Allows switching between pure-CPU to pure-GPU inference
- Supports GGUF (General Graph Universal Format) model format
- Supported Models:
  - Llama3
  - Mistral
  - Phi-3
  - Qwen2.5, 3
  - IBM Granite 3.3, 4.0 (new)
  - Gemma (soon)
- Integration with LangChain4j & Quarkus

Integration with Langchain4j & Quarkus

Integrating GPULlama3.java with LangChain4j in your project

For Maven projects (pom.xml)

```
<dependency>
    <groupId>dev.langchain4j</groupId>
    <artifactId>langchain4j</artifactId>
    <version>1.9.1</version>
</dependency>

<dependency>
    <groupId>dev.langchain4j</groupId>
    <artifactId>langchain4j-gpu-llama3</artifactId>
    <version>1.9.1-beta17</version>
</dependency>
```

For Gradle projects (build.gradle)

```
implementation 'dev.langchain4j:langchain4j:1.9.1'
implementation 'dev.langchain4j:langchain4j-gpu-llama3:1
```

Integrating GPULlama3.java with Quarkus in your project

For Maven projects (pom.xml)

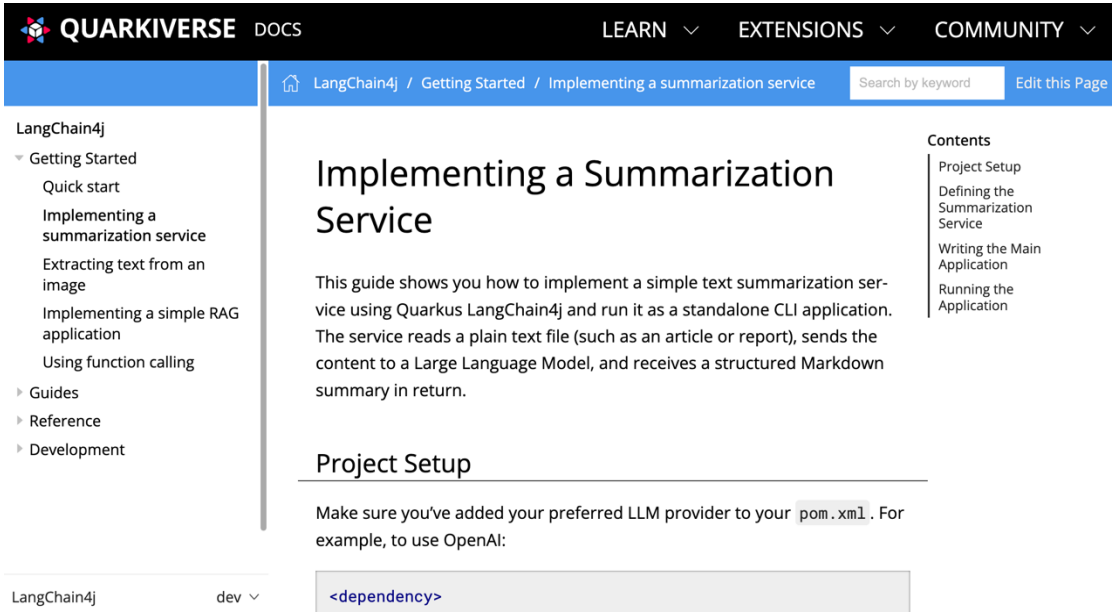
```
<dependency>
    <groupId>io.quarkiverse.langchain4j</groupId>
    <artifactId>quarkus-langchain4j-gpu-llama3</artifac
    <version>1.4.2</version>
</dependency>
```

Integrating GPULlama3.java in your project

For Maven projects (pom.xml)

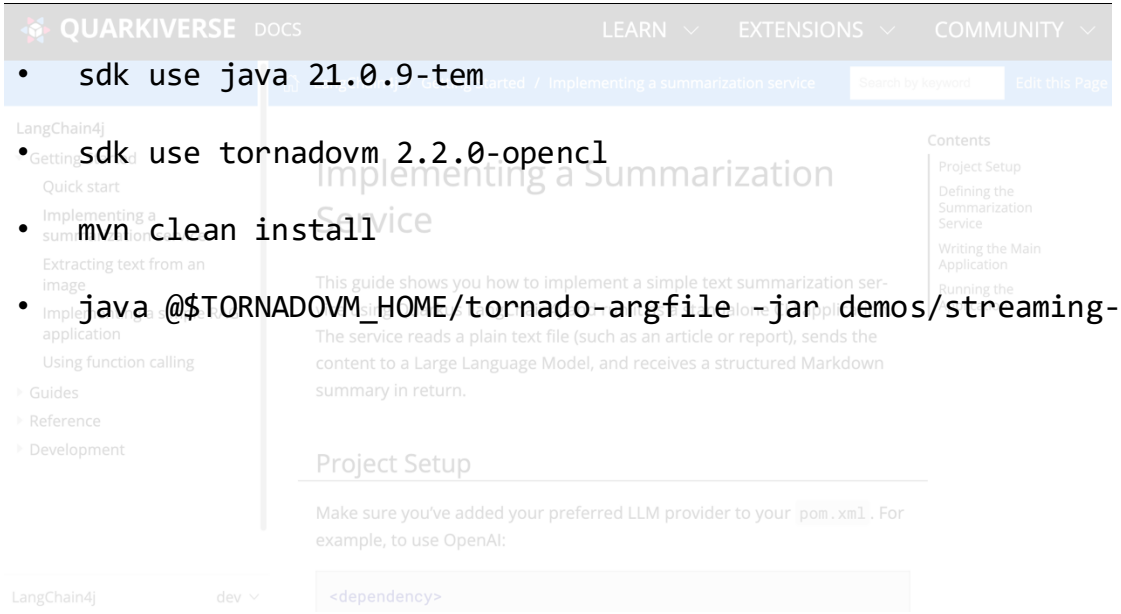
```
<dependency>
    <groupId>io.github.beehive-lab</groupId>
    <artifactId>gpu-llama3</artifactId>
    <version>0.3.3</version>
</dependency>
```

Demo GPULlama3.java, Quarkus, LangChain4j



https://docs.quarkiverse.io/quarkus-langchain4j/dev/quickstart-summarization.html

Demo GPULlama3.java Quarkus, LangChain4j



- `sdk use java 21.0.9-tem`
- `sdk use tornadovm 2.2.0-opencl`
- `mvn clean install`
- `java @$TORNADOVM_HOME/tornado-argfile -jar demos/streaming-demo/target/quarkus-app/quarkus-run.jar`

https://docs.quarkiverse.io/quarkus-langchain4j/dev/quickstart-summarization.html

https://github.com/beehive-lab/gpullama3-quarkus-langchain4j-demo

More about it (Devoxx Belgium 2025)

MANCHESTER 1824
The University of Manchester

DevoxxBE 2025

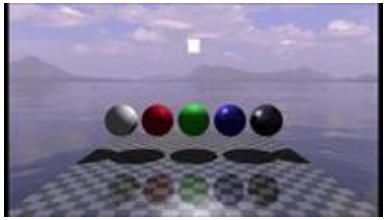
Summary

Use Cases

https://www.tornadovm.org/use-cases

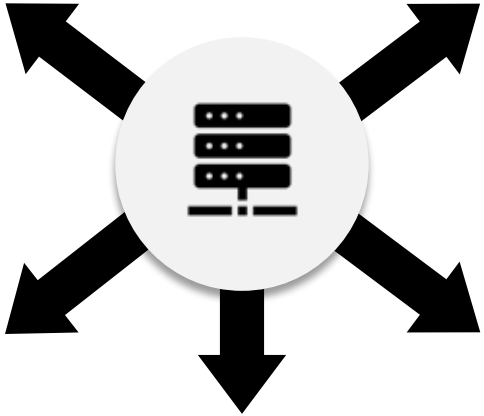
COMPUTER VISION

47x Performance



RAY TRACING

200x Performance



DATA ANALYTICS & NLP

Levenshtein Distance	9.8x
K-means Clustering	9.7x
Hierarchical Clustering	28x

FACE DETECTION

23x Performance

MACHINE LEARNING

14x Performance

Future Roadmap

MANCHESTER 1824
The University of Manchester

- Performance improvements across GPU vendors
- Extended platform support (Metal, AMD)
- AI-specific optimizations and platform integrations
- Compatibility with JDK 25
- Continue successful integration of future Java features
    - Project Babylon, Valhalla, Leyden

Summary

- TornadoVM is an open-source plugin for JDK vendors
- Enables native GPU acceleration of Java programs
- AI-ready (GPULlama3.java, LangChain4J, etc.)
- Production ready
- Rich tool ecosystem
  (Development, Deployment, Performance Monitoring)









Our Team

<u>Academic Staff</u>

- Christos Kotselidis

<u>Research Staff</u>

- Orion Papadakis
- Michail Papadimitriou
- Maria Xekalaki
- Thanos Stratikopoulos
- Ruiqi Ye

<u>Alumni</u>

- James Clarkson
- Foivos Zakkak
- Juan Fumero
- Benjamin Bell
- Amad Aslam
- Ales Kubicek
- Florin Blanaru
- Gyorgy Rethy
- Mihai-Christian Olteanu
- Ian Vaughan
- Tianyu Zuo

Questions?

MANCHESTER 1824
The University of Manchester

https://www.linkedin.com/company/tornadovm

@tornadovm

@tornadovm.org

info@tornadovm.org

https://tornadovm.org

https://github.com/beehive-lab/TornadoVM

https://github.com/beehive-lab/GPULlama3.java

Demo 1 https://github.com/beehive-lab/GPULlama3.java

Demo 2 https://github.com/beehive-lab/gpullama3-quarkus-langchain4j-demo

encrypt AERO P2CODE TANGO oneAPI

UK Research and Innovation